NAME: _____CLASS: _____
STUDENT'S

ST. GODFREY (P.V.T.) SECONDARY SCHOOL

2023 MSCE END OF TERM ONE EXAMINATION

Subject Number: M131/I

Time Allowed: 2 hours8:00 – 10:00 am

PAPER I(100 marks)

PHYSICS

PAPER II (40 MARKS)

INSTRUCTIONS

1. This paper contains **8 printed pages. Please check.**
2. Write your name at the top of each page of the question paper
3. You should spend 30 minutes on each question, the 30 minutes period allowed for each question includes 3 Minute to tidy up the apparatus and have it checked by the supervisor
4. Answer all question in the spaces provided

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SECTION A (20 MARKS)

QUESTION 1 (10 marks)

With the aid of diagrams, explain how the arrangement of two bulbs in a series and a parallel circuit affects brightness of the bulbs.

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QUESTION 2 (10 marks)

A student was investigating how light travels

a) State an appropriate hypothesis for the experiment

b) Design an experiment that could be carried out to test the hypothesis.in the description include; aim of the experiment , materials, procedure of the experiment, results and discussion

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SECTION B (20 Marks)

QUESTION 3 (10 marks)

You are provided with a coin, a pen cover, a small plastic bag, a small stone, water and 4 beakers

a) Pour 40 cm^3 of water in each of the 4 beakers
b) Drop the coin in the first beaker and observe the results
c) Drop the pen cover in the second beaker and observe the results
d) Drop the plastic bag in the third beaker and observe the results
e) Drop the stone in the last beaker and observe the results
f) Record your observations in the table of results

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Table of results

Substance	**Result in water**
Coin	
Pen cover	
Plastic bag	
Stone	

Explain your results

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QUESTION 4 (10 marks)

a) Arrange the apparatus as shown in figure 1

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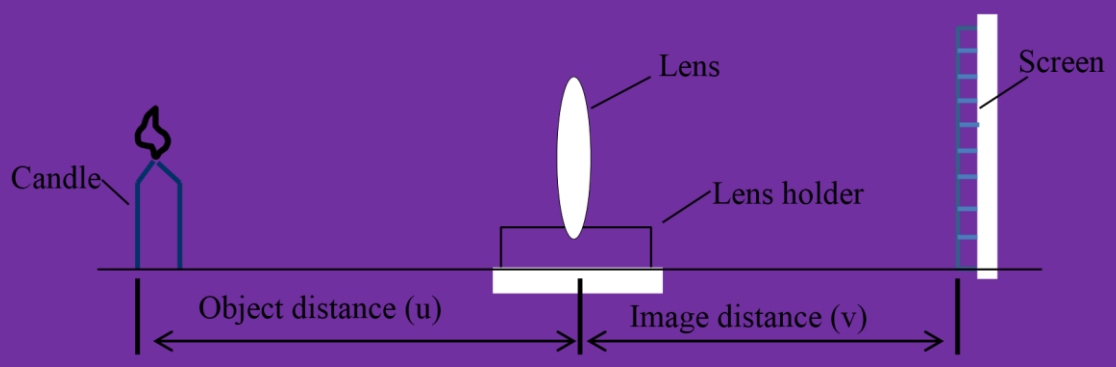

Figure 1

b) Light the candle
c) Move the object(candle)until it is 50cm from the lens
d) Move the screen until a clear Image of the object is formed on the screen
e) Measure the image , V, and record it in the table 1
f) Repeat steps (c) to (e) for the object distances shown in the table

g) Complete the $\frac{1}{v}$ and $\frac{1}{u}$ columns of the table

Object distance, U (cm)	**Image distance V (cm)**	$\frac{1}{v}$	$\frac{1}{u}$
50 40 30 25 20			

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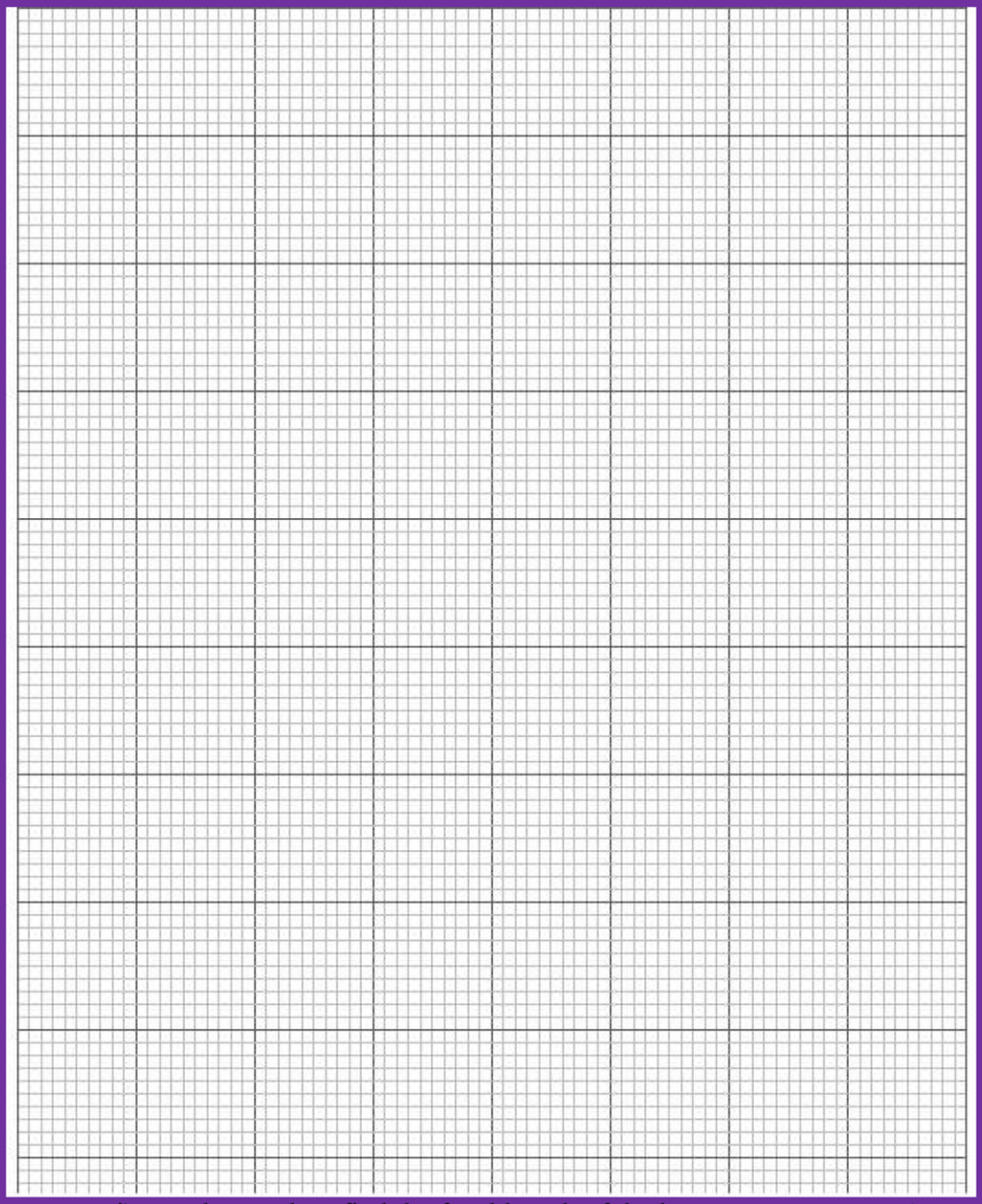
Table 1 **(5 marks)**

h) Plot the graph of − against − **(4 marks)**



i) Use the graph to find the focal length of the lens. **(1 mark)**

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END OF QUESTION PAPER

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